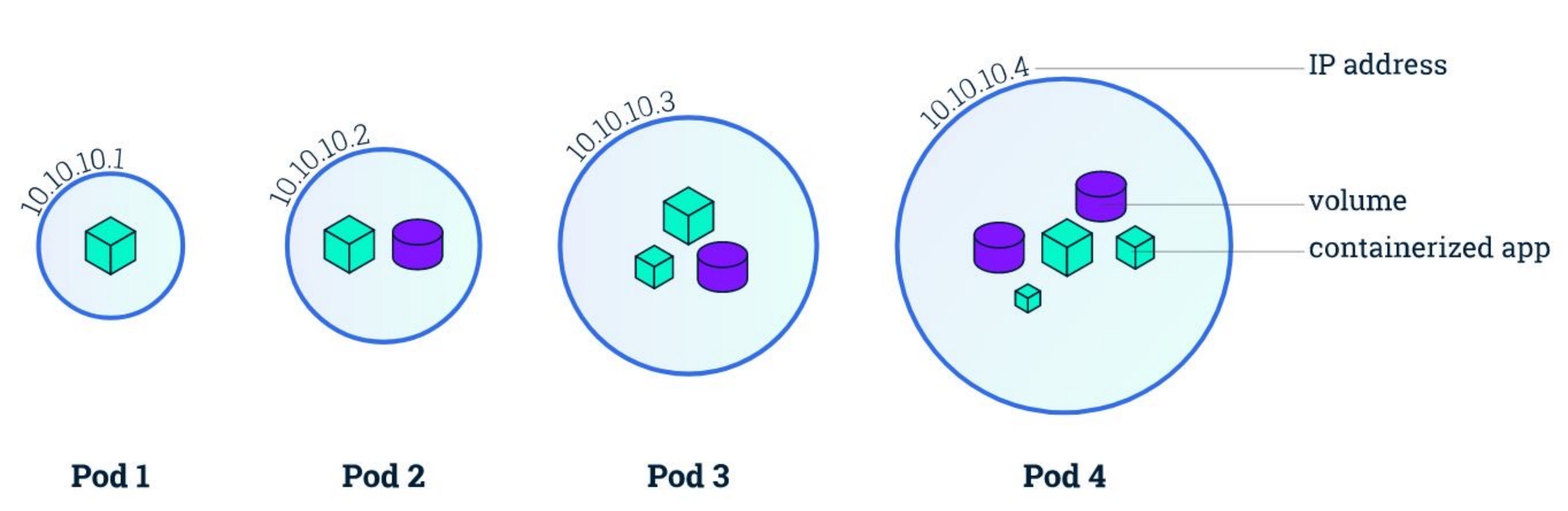
Kubernetes

Pod, Deployment, ReplicaSet and Namespace

Pods

A Pod is the smallest and simplest Kubernetes object. It represents a single instance of a running process in your cluster.

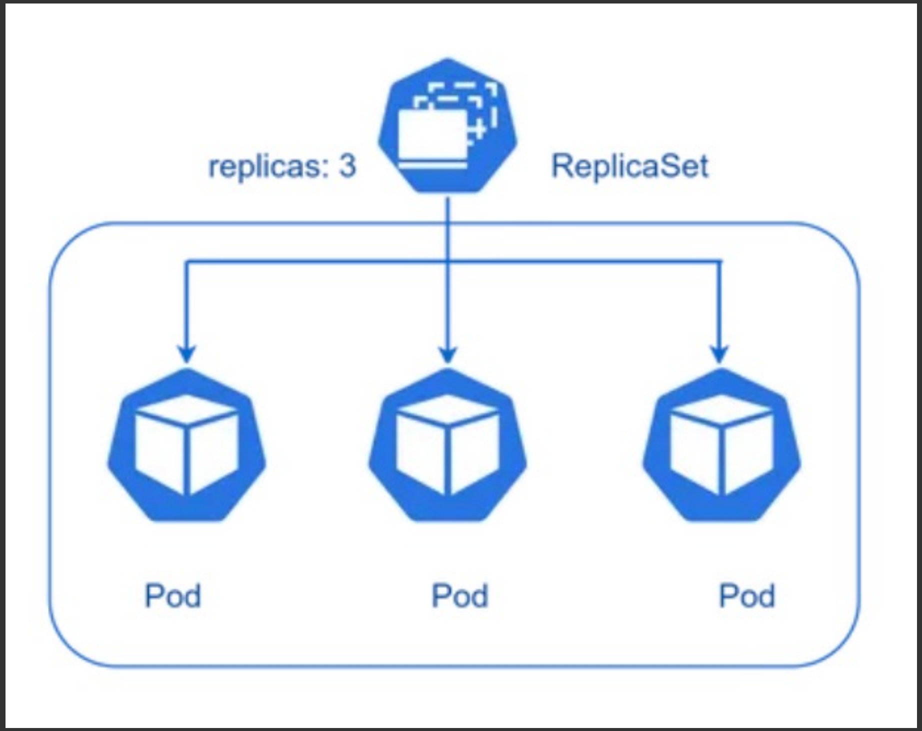
- A Pod can contain one or more containers, which share storage and network resources.
- All containers in a Pod are scheduled on the same node and can communicate with each other via localhost.
- Pods are ephemeral and can be created, destroyed, or replaced dynamically.



ReplicaSet

A ReplicaSet is responsible for maintaining a stable set of replica Pods running at any given time.

- It ensures that a specified number of identical Pods are running, providing high availability.
- If a Pod fails or is terminated, the ReplicaSet automatically creates a new Pod to replace it.
- A ReplicaSet is defined by a label selector that identifies the Pods it manages.



Manage Pods

- kubectl run my-pod --image=nginx
- kubectl get pods: Lists all Pods in the default namespace.
- kubectl get pods -n <namespace>: Lists Pods in a specific namespace.
- kubectl describe pod <pod-name>: Shows detailed information about a specific Pod.
- kubectl logs <pod-name>: Retrieves logs for a specific Pod.
- kubectl exec -it <pod-name> -- /bin/bash: Opens a shell session inside a Pod (if it supports it).
- kubectl delete pod <pod-name>
- kubectl apply -f pod.yaml: Create a Pod (from a YAML file):

Pods and ReplicaSet config

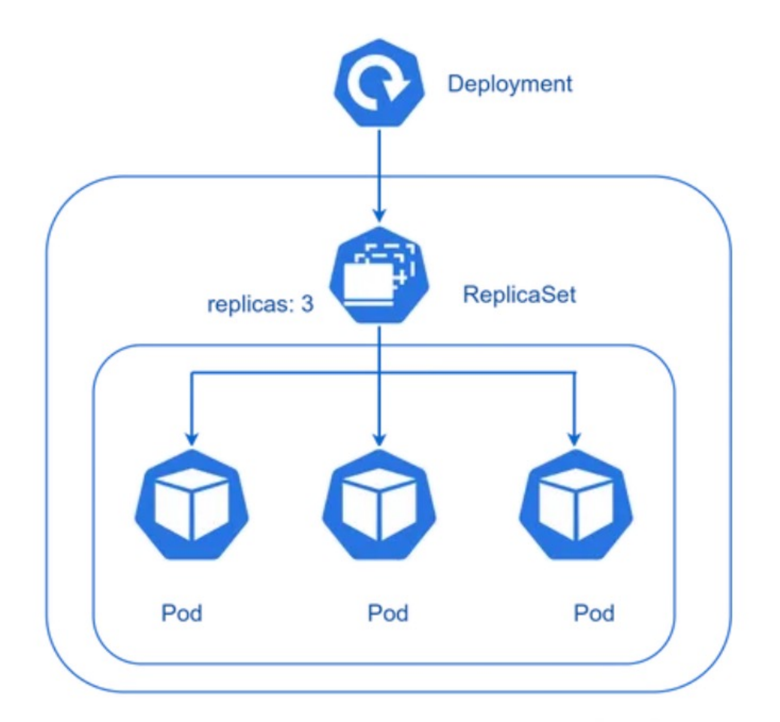
```yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-pod
  labels:
    app: my-app
spec:
  containers:
    - name: my-container
      image: nginx:latest
      ports:
        - containerPort: 80
```

```yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: nginx-replicaset
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Deployment

A Deployment provides declarative updates for Pods and ReplicaSets. It is a higher-level abstraction over ReplicaSets.

- It allows you to describe the desired state of your application (like which images to use, how many replicas, etc.).
- It manages the creation and scaling of ReplicaSets and Pods.
- It enables easy rollbacks to previous versions of your application and handles rolling updates without downtime.



Manage Deployments

- kubectl create deploy <deployment-name> --replicas=3 --image=<image-name>
- kubectl scale deployment <deployment-name> --replicas=5
- kubectl set image deployment/<deployment-name> <container-name>=<new-image> : Update a Deployment
- kubectl rollout undo deployment/<deployment-name> : Rollback to Previous Revision:
- kubectl rollout history deployment/<deployment-name> : Check Rollout History:
- kubectl rollout undo deployment/<deployment-name> --to-revision=<revision-number> : To rollback to a specific revision

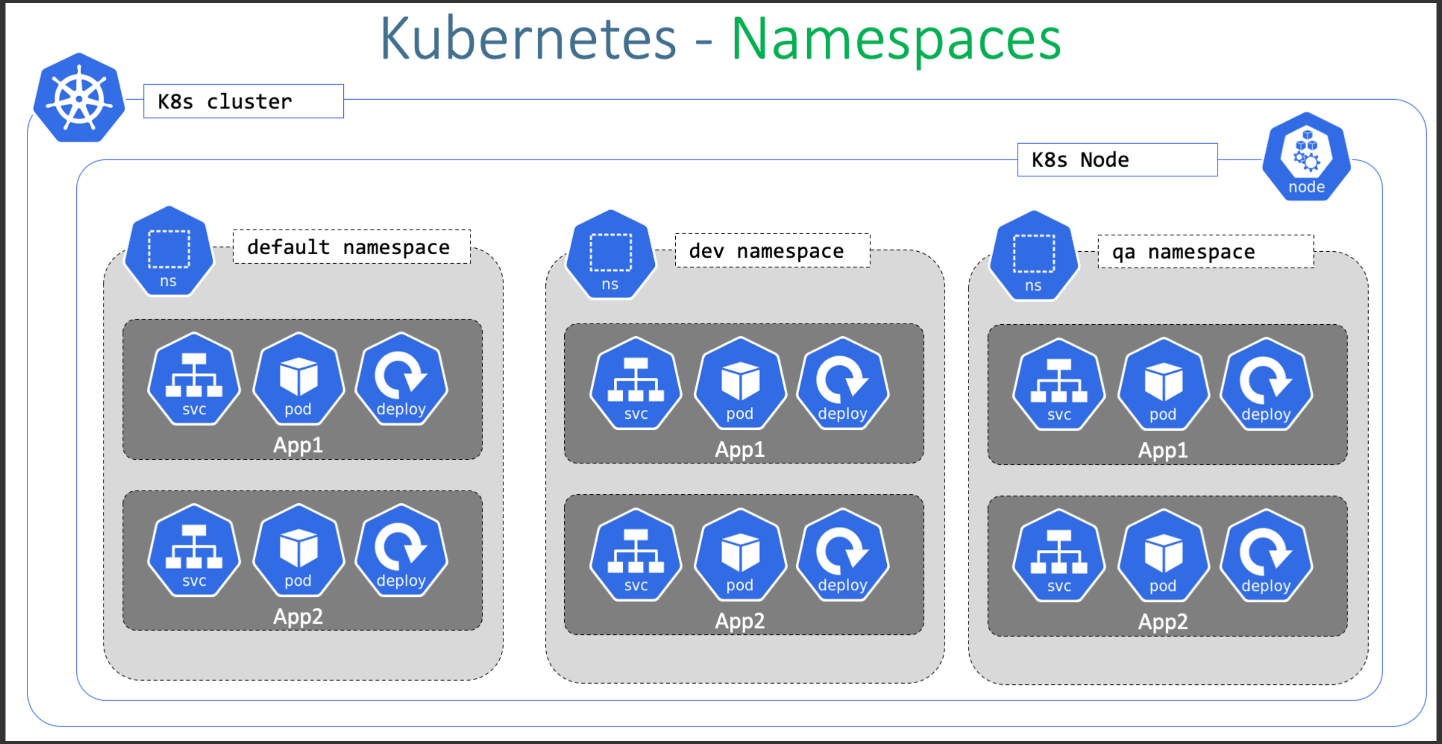
Declarative Configuration

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 3  # Number of replicas
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
        - name: my-app-container
          image: my-app:latest  # Container image
          ports:
            - containerPort: 80
```

```
kubectl apply -f my-app-deployment.yaml
```

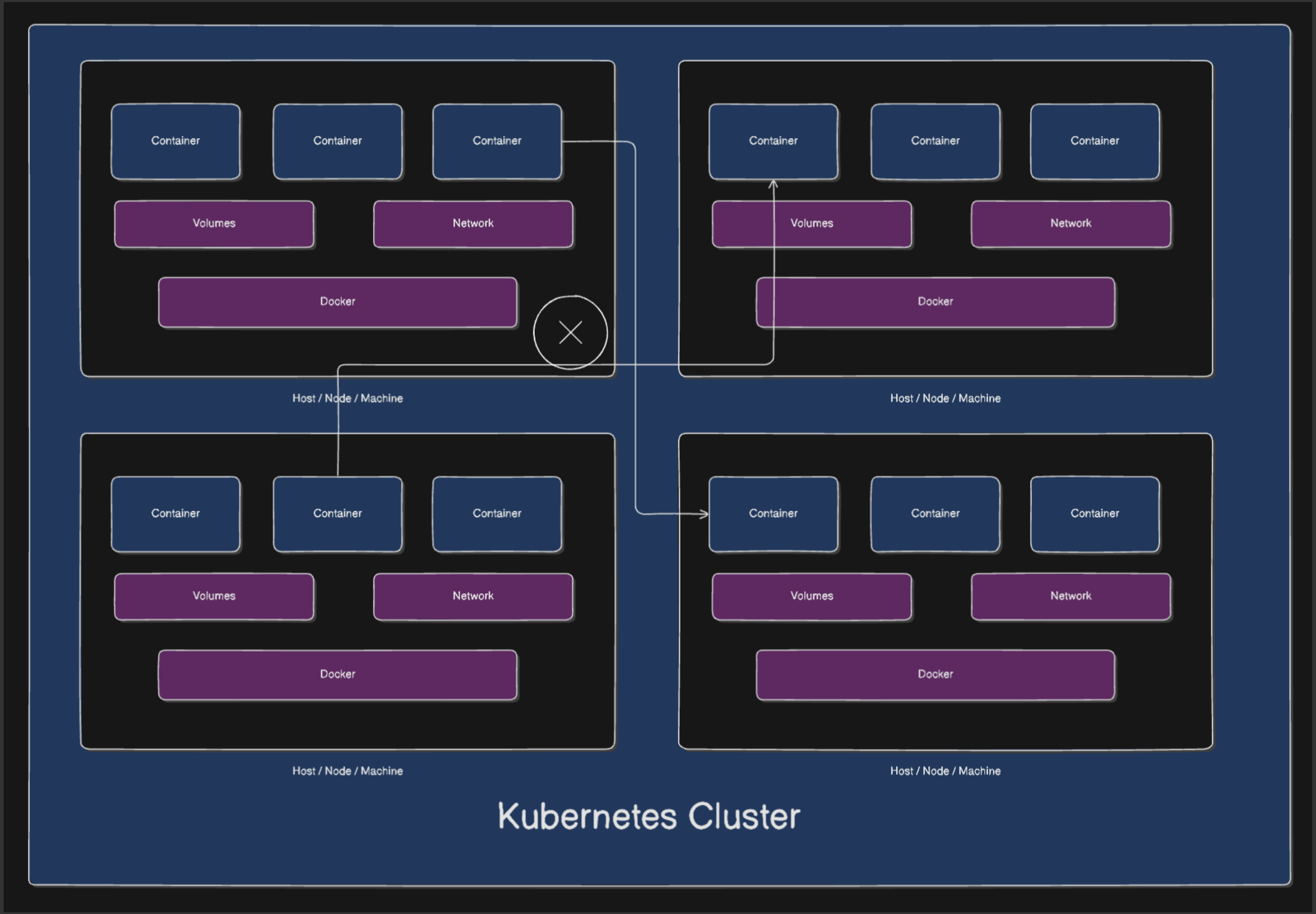
Namespace

A namespace is a logical partitioning of cluster resources that allows for the organization and separation of resources within the cluster. Namespaces provide a way to create multiple virtual clusters within a single physical cluster, allowing teams or applications to operate independently and avoid resource conflicts.



Manage Namespaces

```
kubectl get namespaces or kubectl get ns
kubectl create namespace <namespace-name>
kubectl delete namespace <namespace-name>
kubectl get pods -n <namespace-name>
kubectl config set-context --current --namespace=<namespace-name>
kubectl get pods --all-namespaces or kubectl get pods -A
kubectl describe namespace <namespace-name>
kubectl edit namespace <namespace-name>
kubectl get all --all-namespaces (to get all resources from all namespces)
```


Container
Container
Container
Volumes
Network
Docker
Host / Node / Machine
Container
Container
Container
Volumes
Network
Docker
Host / Node / Machine
Container
Container
Container
Volumes
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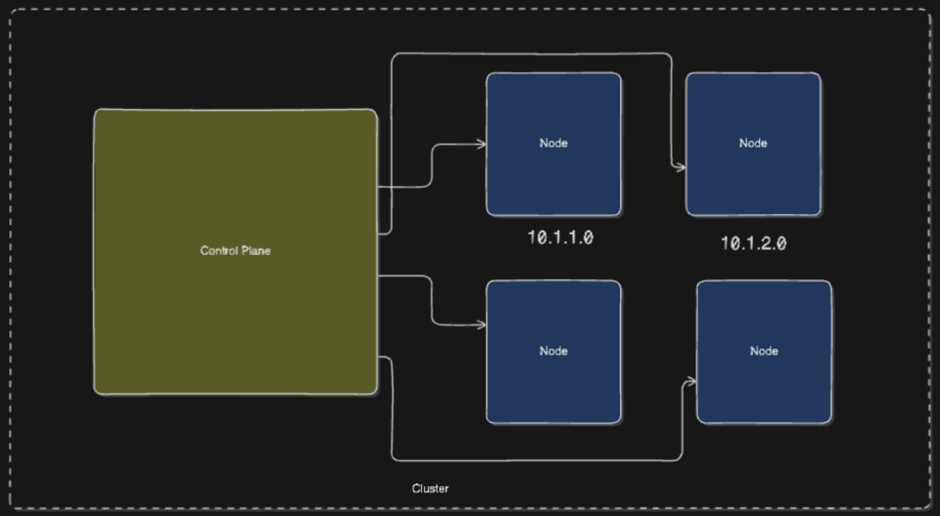
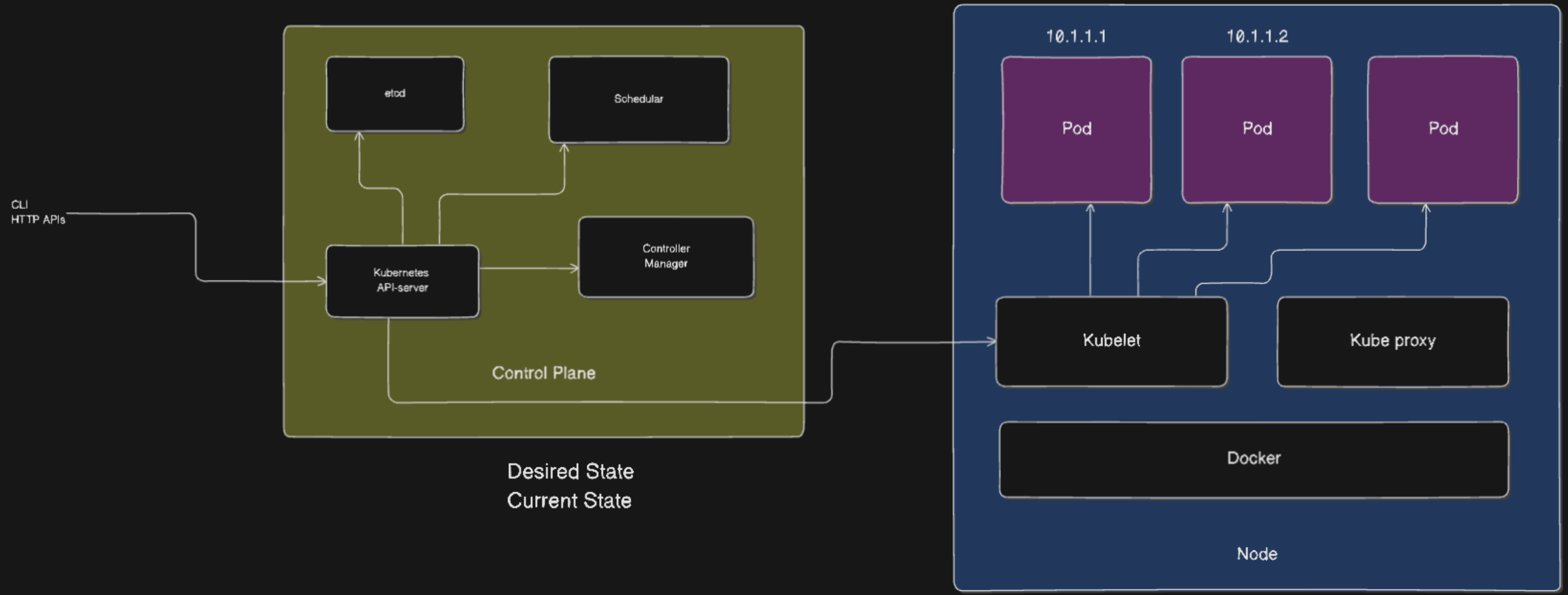
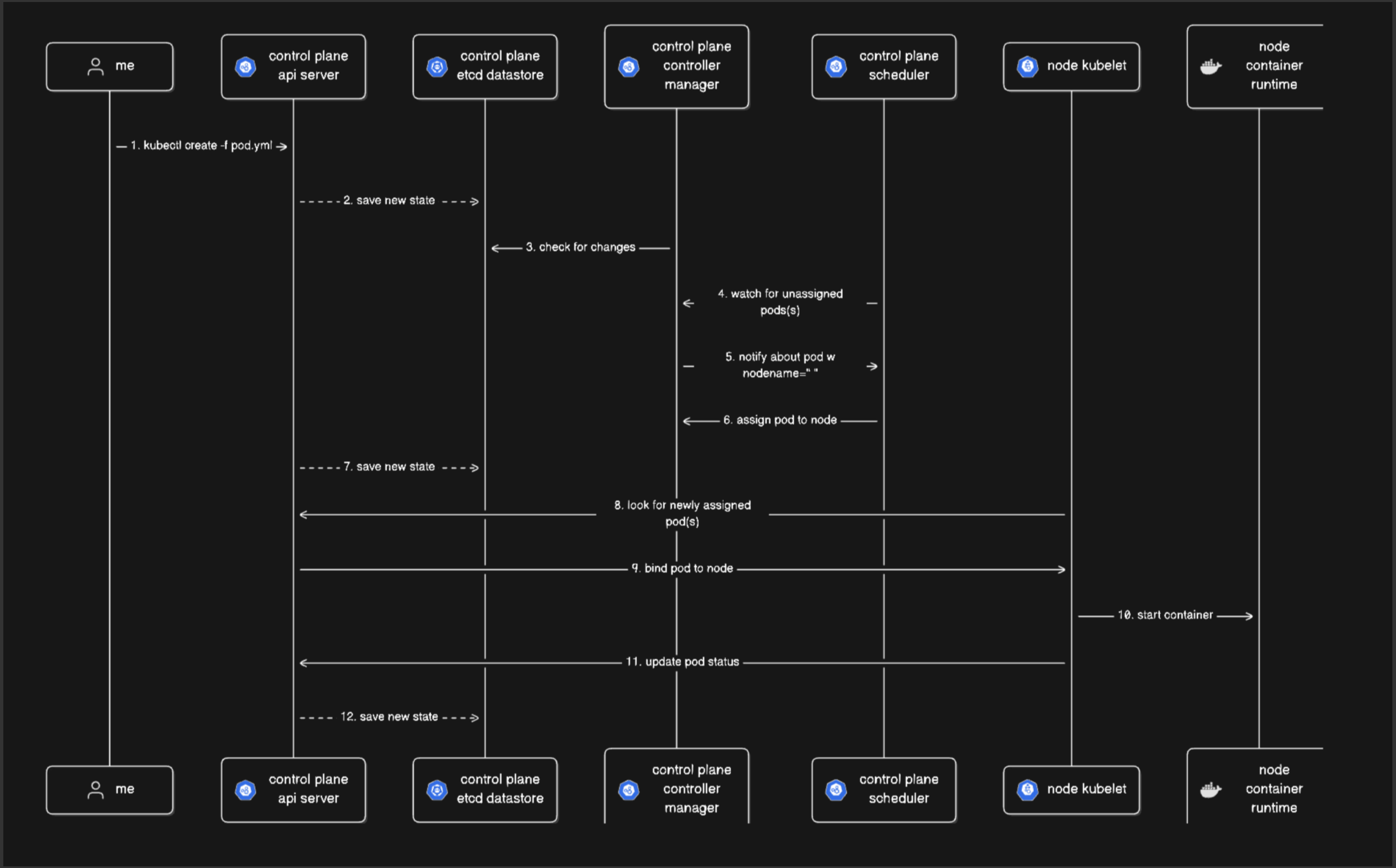
Host / Node / Machine
Container
Container
Container
Volumes
Network
Docker
Host / Node / Machine
Kubernetes Cluster

CODING SHUTTLE







CODING SHUTTLE